

Yukai Jin
jineric73@gmail.com
ericjin73.github.io

Education

2023-26 **Doctor of Philosophy**, Hiroshima University, Japan
2020-23 **Master in Engineering**, Guangdong University of Technology, China
2016-20 **Bachelor in Engineering**, Jiangxi Agricultural University, China

Employment

2026– **Researcher (Specially Appointed)**, National Institute for Environmental Studies (NIES), Japan

Research Experience

2025 Visiting Assistant Research, Yale University, (Advisor: [Prof. Karen Seto](#))
2024 Research Assistant, Hiroshima University, (Advisor: [Prof. Ayyoob Sharifi](#))
2024-25 Teaching Assistant, Hiroshima University

Awards & Honors

2026 HU Student Award, Hiroshima University
2025 Hiroshima University Excellent Student Scholarship
2025 HU Spring Award (JPY¥ 120,000)
2024-26 New Spring Fellowship (JPY¥ 5,700,000)
2023-24 SmaSo-X Scholarship (JPY¥ 150,000)
2022 Graduate Scholarship First (CNY¥ 10,000)
2020 Graduate Scholarship Third (CNY¥ 6,000)

Projects

1. **Jin, Yukai & Shao, Y.** *Predicting CO₂ Emission Reductions of Electric Vehicles in European Cities*, HU Spring Research Project, Japan Science and Technology Agency (JST), Japan. 2025.
– Amount: JPY¥ 1,000,000
– Roll: PI.
2. **Jin, Yukai.** *Predicting CO₂ emissions for global cities from 2021 to 2100*, HU Spring Study Abroad Project, Japan Science and Technology Agency (JST), Japan. 2024.
– Amount: JPY¥ 1,000,000.
3. **Jin, Yukai.** *A deep learning approach to analyze future urban emission pathways*, SmaSo-X Challenge Project Research Grant, Hiroshima University, Japan. 2023.
– Amount: JPY¥ 200,000.

Publications

Journal Articles

1. **Jin, Yukai** & Sharifi, A. Diagnosing systematic overestimation and feature dependencies in data-driven predictions of urban CO₂ emissions in China. *Environmental Impact Assessment Review* **123**. <https://doi.org/10.1016/j.eiar.2026.108770> (2027).
– Impact Factor = 12.2 (Q1).
2. **Jin, Yukai** & Sharifi, A. Estimated municipality-level electric vehicle and passenger vehicle fleet composition across 14 European countries, 2011–2024. *Scientific Data*. <https://doi.org/10.1038/s41597-026-08445-z> (2026).
– Impact Factor = 7.2 (Q1)
– In press.
3. **Jin, Yukai** & Sharifi, A. Heterogeneous urban decarbonization pathways revealed by typological and machine learning frameworks: A global analysis across Annex I cities. *Urban Climate* **70**. <https://doi.org/10.1016/j.uclim.2026.103160> (2026).
– Impact Factor = 7.5 (Q1).
4. **Jin, Yukai** & Sharifi, A. Machine learning for predicting urban greenhouse gas emissions: A systematic literature review. *Renewable and Sustainable Energy Reviews* **215**. <https://doi.org/10.1016/j.rser.2025.115625> (2025).
– Impact Factor = 18.0 (Q1).
5. Sharifi, A., **Jin, Yukai** & Aboagye, P. D. Mapping and topic modeling of over 133,000 studies on climate change and cities: Trends, geographical disparities, and future research needs. *Urban Climate* **61**. <https://doi.org/10.1016/j.uclim.2025.102448> (2025).
– Impact Factor = 7.5 (Q1).
6. **Jin, Yukai**, Sharifi, A., Li, Z., Chen, S., Chen, S. & Zhao, S. Analyzing particulate matter characteristics of the subway system: Case study of Guangzhou. *Case Studies in Chemical and Environmental Engineering* **10**. <https://doi.org/10.1016/j.cscee.2024.100953> (2024).
7. **Jin, Yukai**, Sharifi, A., Li, Z., Chen, S., Zeng, S. & Zhao, S. Carbon emission prediction models: A review. *Science of The Total Environment* **927**. <https://doi.org/10.1016/j.scitotenv.2024.172319> (2024).
– Highly Cited Paper.
8. Li, Z., **Jin, Yukai**, Liang, X. & Zeng, J. Thermography evaluation of defect characteristics of building envelopes in urban villages in Guangzhou, China. *Case Studies in Construction Materials* **17**. <https://doi.org/10.1016/j.cscm.2022.e01373> (2022).
– Impact Factor = 6.9 (Q1).

Peer-reviewed Conference Proceedings

1. **Jin, Yukai** & Sharifi, A. Predicting long-term building energy consumption using multiple feature clustering and machine learning: applications in Shanghai, China in *Proceedings of the 16th International Conference on Applied Energy*, 2024, Niigata, Japan **54** (2024). <https://doi.org/10.46855/energy-proceedings-11538>.

Presentations

1. **Jin, Yukai.** *Predicting Electric Vehicles Emission Reduction Performance in Nordic Cities*, 5th The Network for Education and Research on Peace and Sustainability (NERPS 2026), Tokyo, Japan. 2026.
2. **Jin, Yukai.** *Predicting long-term building energy consumption using multiple feature clustering and machine learning: applications in Shanghai, China*, 16th International Conference on Applied Energy (ICAE2024), Niigata, Japan. 2024.

Teaching

Hiroshima University

2025 Teaching Assistant, Urban Policy (CODE: WMH01901)
2024-2025 Teaching Assistant, Public Administration and Management (CODE: WMH01501)

Guangzhou College of Technology and Business

2022 Part-time Lecturer, Structural Mechanics

Academic Service

Journal Reviewer

2025 International Journal of Digital Earth (Q1)
2026 City and Environment Interactions (Q2)
2026 Scientific Reports (Q2)
2026 Water, Air, & Soil Pollution (Q2)
2026 Modeling Earth Systems and Environment (Q3)
2026 Journal of Applied Meteorology and Climatology (Q3)

Last updated: October 10, 2026